
Education

- 2023–2027 (expected) **PhD in Geology**, *University of Colorado, Boulder*.
- 2018–2020 **MS in Mathematics**, *University of Vermont*, GPA: 4.00.
Extremal/Saturation Numbers and an Entropy Result for the Guessing Numbers of Undirected Graphs
- Advisor: Puck Rombach, Ph.D.
- 2012–2016 **BA in Geology**, *Oberlin College*, GPA: 3.73, Major GPA: 3.91.
Honors: Phi Beta Kappa, Sigma Xi

Experience

Research

- August 2023–Current **Interpreting Convolutional Neural Networks to Understand the Relationship Between Topography and Groundwater**, *University of Colorado, Boulder, CO*, Supervisor: Gregory Tucker, Ph.D..
- August 2019–June 2022 **Guessing Numbers of Graphs**, *University of Vermont, Burlington, VT*, Supervisor: Puck Rombach, Ph.D..
- Summer 2015 **Geologic Mapping of Putative Paleolake Deposits in a Coprates Catena, Mars**,
January 2016 *Laboratory for Atmospheric and Space Physics, Boulder, CO*, Supervisor: Brian Hynek, Ph.D..
- June 2014–August 2016 **Undergraduate Research in Geomorphology**, *Oberlin College, Oberlin, OH*, Supervisor: Amanda Schmidt, Ph.D..
Oberlin College
- January 2015 **Paleomagnetism of the Samial Ophiolite in Oman**, *Oberlin College, Oberlin, OH*, Supervisor: Andrew Horst, Ph.D..

Teaching

- 2023–2024 **Graduate Teaching Assistant**, *University of Colorado, Boulder, CO*.
 - GEOL 1030 (Introduction to Geology Laboratory), Spring 2024: independently instructed 2 lab sections.
 - GEOL 3600 (Introduction to Python Programming for Earth Scientists), Fall 2023
- 2018–2020 **Graduate Teaching Assistant**, *University of Vermont, Burlington, VT*.
 - GEOL 007 (Earth Hazards), Spring 2020
 - GEOL 001 (Introductory Geology), Fall 2018, Spring 2019, Fall 2019: independently instructed 2–3 lab sections.
- 2016–2018 **Instructor**, *Oberlin College, Oberlin, OH*, .
 - GIS for Natural and Social Sciences, Winter 2017, Winter 2018
 - The Art of Data Visualization, Spring 2016, Spring 2017, Spring 2018
- 2016–2018 **Teaching Assistant**, *Oberlin College, Oberlin, OH*, .
 - GEOL 235 (Applied GIS), Fall 2016, Fall 2017
- 2014 **Undergraduate Teaching Assistant**, *Oberlin College, Oberlin, OH*, .
 - GEOL 235 (Applied GIS), Fall 2014
 - GIS for Natural and Social Sciences, January 2014

- 2016 **Tutor**, *Oberlin College*, Oberlin, OH, .
- Applied GIS, Fall 2015
 - Multivariable Calculus, Spring 2016
 - Earth Surface Processes, Spring 2016
- Fall 2013 **Computer Science Lab Helper**, *Oberlin College*, Oberlin, OH, .
- CSCI 150 & 151 (Intro to Computer Science and Algorithms)
- Nonacademic Employment
- 2020–2023 **Security Engineer**, *NuHarbor Security*, Colchester, VT, .

Journal Publications

J. R. Martin and P. Rombach, "Guessing numbers and extremal graph theory," *The Electronic Journal of Combinatorics*, vol. 29, no. 2, Jun. 2022. DOI: 10.37236/9851. [Online]. Available: <https://doi.org/10.37236/9851>.

D. Pennington, I. Ebert-Uphoff, N. Freed, **J. R. Martin**, and S. A. Pierce, "Bridging sustainability science, earth science, and data science through interdisciplinary education," *Sustainability Science*, pp. 1–15, 2019.

Abstracts

A. J. Schmidt, P. R. Bierman, V. Sosa-Gonzalez, T. B. Neilson, D. H. Rood, **J. R. Martin**, and M. Hill, "Agricultural land use doubled sediment yield in western China's rivers," in *AGU Fall Meeting Abstracts*, 2017.

A. J. Horst, S. Trutner, E. S. Hartley, and **J. R. Martin**, "New paleomagnetic data from upper gabbros supports limited rotation of central Semail massif in Oman ophiolite," in *AGU Fall Meeting Abstracts*, 2017.

J. R. Martin and A. H. Schmidt, "Teaching python scripting through arcgis," in *Geological Society of America Abstracts with Programs*, vol. 49, 2017.

A. J. Horst, S. Trutner, E. S. Hartley, and **J. R. Martin**, "Development of foliations in uppermost gabbros in the Oman ophiolite and implications for crustal accretion along a paleoridge axis," in *Geological Society of America Abstracts with Programs*, vol. 48, 2016.

J. R. Martin, B. M. Hynek, and M. Chojnacki, "Geologic mapping of putative paleolake deposits in a Coprates catena, Mars," in *Lunar and Planetary Science Conference*, vol. 47, 2016, p. 2625.

M. Chojnacki, B. M. Hynek, S. R. Black, R. Hoover, and **J. R. Martin**, "Geologic mapping of the Coprates Chasma (MTM-15057), Mars: Year 2," in *Lunar and Planetary Science Conference*, vol. 47, 2016, p. 2828.

J. R. Martin, D. Lawrence, A. H. Schmidt, P. Bierman, V. Sosa-Gonzalez, A. Singleton, and Y. Qui, "Examining land use and erosion in southwest China," in *Geological Society of America Abstracts with Programs*, vol. 47, 2015, p. 515.

B. M. Hynek, M. Chojnacki, S. R. Black, and **J. R. Martin**, "Geologic mapping of the Coprates Chasma (MTM-15057), Mars: Year 1," in *Abstracts of the Annual Meeting of Planetary Geologic Mappers*, 2015.

G. Garcia Greco, D. Fialo, T. B. Neilson, **J. R. Martin**, R. Wei, C. Zhang, and C. Liang, "Determining sediment sources and sinks in a small headwater watershed in Yunnan, China using short-lived radionuclides," in *Geological Society of America Abstracts with Programs*, vol. 46, 2014, p. 800.

J. R. Martin and G. E. Tucker, "Inferring geomorphic model parameters from topography using convolutional neural networks," *AGU24*, 2024.

Other Publications

J. R. Martin and A. H. Schmidt, "Lab 3: Changes to the green river," in *On the Cutting Edge Peer Reviewed Teaching Activities Collection*, 2017.

——, "Lab 4: Hurricanes," in *On the Cutting Edge Peer Reviewed Activities Teaching Collection*, 2017.

——, "Lab 5: Sea level rise," in *On the Cutting Edge Peer Reviewed Activities Teaching Collection*, 2017.

——, "Differences and similarities between mean local relief and slope," in *On the Cutting Edge Peer Reviewed Activities Teaching Collection*, 2017.

——, "Lab 7: Hawai'i," in *On the Cutting Edge Peer Reviewed Activities Teaching Collection*, 2017.

——, "Lab 8: Lidar," in *On the Cutting Edge Peer Reviewed Activities Teaching Collection*, 2017.

Conference Activity

Sessions Organized

- 2025 **19th Annual Hydrologic Sciences Student Symposium**, Boulder, CO, 10-11 April 2025.
Steering Committee Member
- 2024 **18th Annual Hydrologic Sciences Student Symposium**, Boulder, CO, 11-12 April 2024.
Steering Committee Member
- 2018 **Educating the Next Generation of Scientists: Interdisciplinary Education Bridging Geoscience and Data Science**, *American Geophysical Union Fall Meeting 2018*, Washington, DC, 14 December 2018.
Co-convended a session of talks and a session of posters.

Talks Given

- 2025 **Convolutional Neural Networks and the Topographic Fingerprint of Water**, *CU Boulder HydroSciences Student Symposium*, Boulder, CO, 11 April 2025.
- 2017 **Teaching Python Scripting Through ArcGIS**, *Geological Society of America Northeast/North central Section Meeting*, Pittsburgh, PA, 19 March 2017.

Posters Presented

- 2024 **Neural Networks Can Infer Geomorphic Model Parameters From Topography**, *American Geophysical Union 2024 Fall Meeting*, Washington, DC, 10 December 2024.
- 2024 **Measurement and Threshold Sensitivities in the Fractal Dimension of River Networks**, *CSDMS 2024 Annual Meeting*, Montclair, NJ, 15 May 2024.
- 2018 **IS-GEO - A Research Coordination Network on Intelligent Systems Research to Support the Geosciences**, *EarthCube 2018 All Hands Meeting*, Alexandria, VA, 7 June 2018.
- 2017 **Teaching Python Scripting to the Next Generation of Geoscientists**, *EarthCube 2017 All Hands Meeting*, Seattle, WA, 8 June 2017.
- 2016 **Geologic mapping of putative paleolake deposits in a Coprates catena, Mars**, *Lunar and Planetary Science Conference*, The Woodlands, TX, 22 March 2016.

- 2015 **Examining land use and erosion in southwest China**, *Geological Society of America Annual Meeting*, Baltimore, MD, 3 November 2015.

Workshops

- August 2018 **2018 IS-GEO Summer Institute**, *University of Hawai'i at Hilo*, Hilo, HI.
July 2017 **2017 IS-GEO Summer Institute**, *Texas Advanced Computing Center*, Austin, TX.

Service

- 2024–2025 **Secretary of the Geological Sciences Graduate Student Association**, *CU Boulder*, Boulder, CO.
2024, 2025 **Steering Committee Member for the Hydrologic Sciences Student Symposium**, *CU Boulder*, Boulder, CO.
2017–2019 **Education Committee Member**, *IS-GEO Research Coordination Network*.

Awards and Honors

- 2024 **Harlan Erker Memorial Scholarship**, *Colorado Groundwater Association*, Conifer, CO.
2024 **Timothy William Stanton Scholarship**, *Department of Geology*, University of Colorado, Boulder, CO.
2023 **Chancellor's Fellowship**, University of Colorado, Boulder, CO.
2021 **2020-2021 Outstanding Master's Thesis Award**, *Graduate College*, University of Vermont, Burlington, VT.
2020 **John F. Kenney Award for Mathematics**, *Department of Mathematics*, University of Vermont, Burlington, VT.
2020 **Geology Graduate Teaching Assistant Award**, *Department of Geology*, University of Vermont, Burlington, VT.
2016 **George B. Wharton Prize in Geology**, *Department of Geology*, Oberlin College, Oberlin, OH.